

LAKSHYA

JEE 2027

Lecture No.- 20

MATHEMATICS

RELATIONS & FUNCTIONS

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Topics *to be covered*



1 Equal & Identical Functions

2 Classification of Functions

3 Question Practice

4





Question Practice

Question



Find range using derivative:

- (i) $f(x) = x^4 - 3x^2 + 1$, $x \in (-\infty, \infty)$
- (ii) $f(x) = x^4 - 3x^2 + 1$, $x \in [-2, 3]$
- (iii) $f(x) = 4x^3 + x^2 + 7x + 1$, $x \in [-1, 2)$

$$f(0) = 1, \quad f\left(\pm\sqrt{\frac{3}{2}}\right) = -\frac{5}{4}$$
$$f(-2) = 5$$
$$f(3) = 55$$
$$\text{Range} \in \left[-\frac{5}{4}, 55\right]$$

$$y = x^4 - 3x^2 + 1$$
$$\frac{dy}{dx} = 4x^3 - 6x = 0$$
$$2x(2x^2 - 3) = 0$$
$$\# x = 0, x = \pm\sqrt{\frac{3}{2}}$$

$$\# f(0) = 1$$

$$\# f\left(\pm\sqrt{\frac{3}{2}}\right) = -\frac{5}{4}$$

$$\# f(\infty) \Rightarrow \text{as } x \rightarrow \infty \left(x^4 - 3x^2 + 1 \right) \rightarrow \infty$$

$$\# f(-\infty) \Rightarrow \infty$$

$$\underbrace{x^2}_{\downarrow \infty \times \infty} (\underbrace{x^2}_{\downarrow \infty} - 3) + 1$$

$$\# \text{Range} \in \left[-\frac{5}{4}, \infty\right)$$

Question



Find range using derivative:

(i) $f(x) = x^4 - 3x^2 + 1$, $x \in (-\infty, \infty)$

(ii) $f(x) = x^4 - 3x^2 + 1$, $x \in [-2, 3]$

(iii) $f(x) = 4x^3 + x^2 + 7x + 1$, $x \in [-1, 2]$

$\frac{dy}{dx} = 12x^2 + 2x + 7 = 0$
No Real Root

$\frac{dy}{dx} > 0$, always

min. = $f(-1) = -9$
max. = $f(2) = 51$
Range $\in [-9, 51]$

Question [JEE Mains-2021]



70%

Let $A = [a_{ij}]$ be a 3×3 matrix, where $a_{ij} = \begin{cases} 1, & \text{if } i = j \\ -x, & \text{if } |i - j| = 1 \\ 2x + 1, & \text{otherwise} \end{cases}$

Let a function $f: R \rightarrow R$ be defined as $f(x) = \det(A)$. Then the sum of maximum and minimum values of f on R is equal to

☒ **A** $-\frac{88}{27}$

☐ **B** $\frac{20}{27}$

☐ **C** $\frac{88}{27}$

☐ **D** $-\frac{20}{27}$

$$A = \begin{bmatrix} 1 & -x & 2x+1 \\ -x & 1 & -x \\ 2x+1 & -x & 1 \end{bmatrix}_{3 \times 3}$$

$|A| = \checkmark$

$f(x) = |A|$

Question [JEE Mains-2025]

75%



Let $f: [0, 3] \rightarrow A$ be defined by $f(x) = 2x^3 - 15x^2 + 36x + 7$ and $g: [0, \infty) \rightarrow B$ be defined by $g(x) = \frac{x^{2025}}{x^{2025} + 1}$. If both the functions have Range equals to codomain and $S = \{x \in \mathbb{Z} : x \in A \text{ or } x \in B\}$, then $n(S)$ is equal to

A 36

B 29

☒ C 30

D 31

$$f(x) = 2x^3 - 15x^2 + 36x + 7$$

$$\frac{dy}{dx} = 6x^2 - 30x + 36 = 0$$

$$6(x^2 - 5x + 6) = 0$$

$$\# \boxed{x = 2, 3}$$

$$f(2) = 35$$

$$f(3) = 34$$

$$f(0) = 7$$

$$\text{Range} \in \overset{A}{\boxed{[7, 35]}}$$

$$g(x) = 1 - \frac{1}{x^{2025} + 1}, \quad x \in [0, \infty)$$

$$x^{2025} \in [0, \infty)$$

$$x^{2025} + 1 \in [1, \infty)$$

$$1 - \frac{1}{x^{2025} + 1} \in \boxed{[0, 1)}$$

$\underset{B}{}$

Acidic Question



Find range of: $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}, x \in [-1, 3]$

$$\left[\frac{1}{3}, 3\right]$$

$$\frac{dy}{dx} = \frac{(x^2 + x + 1)(2x - 1) - (x^2 - x + 1)(2x + 1)}{(x^2 + x + 1)^2} = 0$$

$$\frac{u}{v} \xrightarrow{d} \frac{v \cdot u' - u \cdot v'}{v^2}$$

$$f(-1) = 3 \checkmark$$

$$f(1) = \frac{1}{3} \checkmark$$

$$f(3) = \frac{7}{13}$$

$$(x^2 + x + 1)(2x - 1) = (x^2 - x + 1)(2x + 1)$$

$$\cancel{2x^3} + \cancel{2x^2} + \cancel{2x} - \cancel{x^2} - \cancel{x} - 1 = \cancel{2x^3} - \cancel{2x^2} + \cancel{2x} + \cancel{x^2} - \cancel{x} + 1$$

$$x^2 - 1 = -x^2 + 1$$

$$2x^2 - 2 = 0$$

$$x = \pm 1$$

Acidic Question



Find range of: $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}, x \in [-1, 3]$



f $\underbrace{[-1, 3]}_{\text{dom}} \rightarrow \underbrace{\left[\frac{1}{3}, 3\right]}_{\text{cod}}$, $f(x) = \underline{\hspace{2cm}} \in \left[\frac{1}{3}, 4\right]$



SACHIN SIR SPECIAL SUMMARY(S⁴) FOR RANGE



Range of Functions Involving GIF & FPF:

Format: $f(x)$ = any function of GIF & FPF.

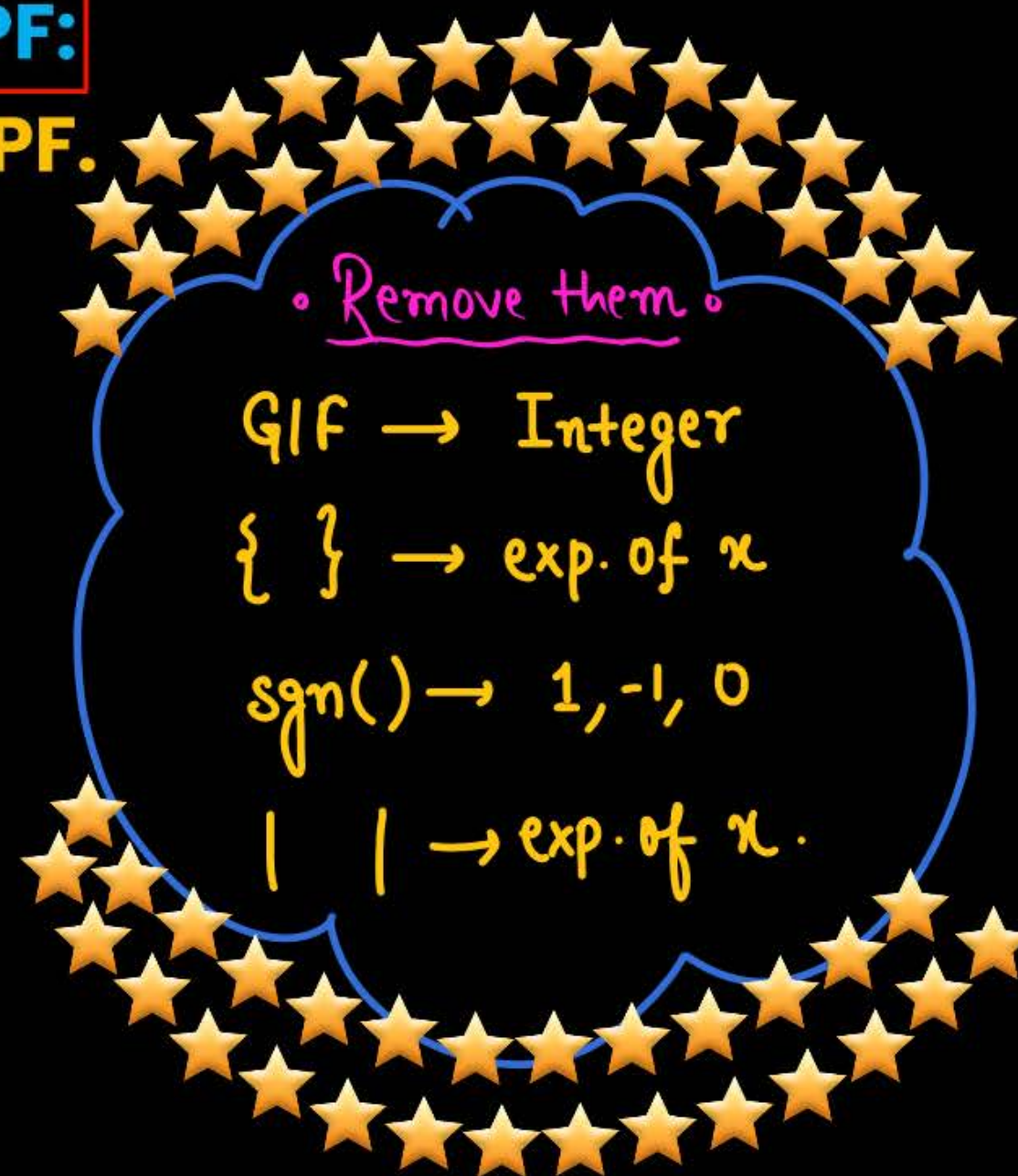
Examples:

$$f(x) = [\sin x]$$

$$f(x) = [x^2], x \in [0, 3]$$

$$f(x) = \left[\frac{2}{x} \right], x \in [3, 4)$$

$$f(x) = x + \{x\}, x \in [-1, 2)$$





Question

Find range of following Functions:

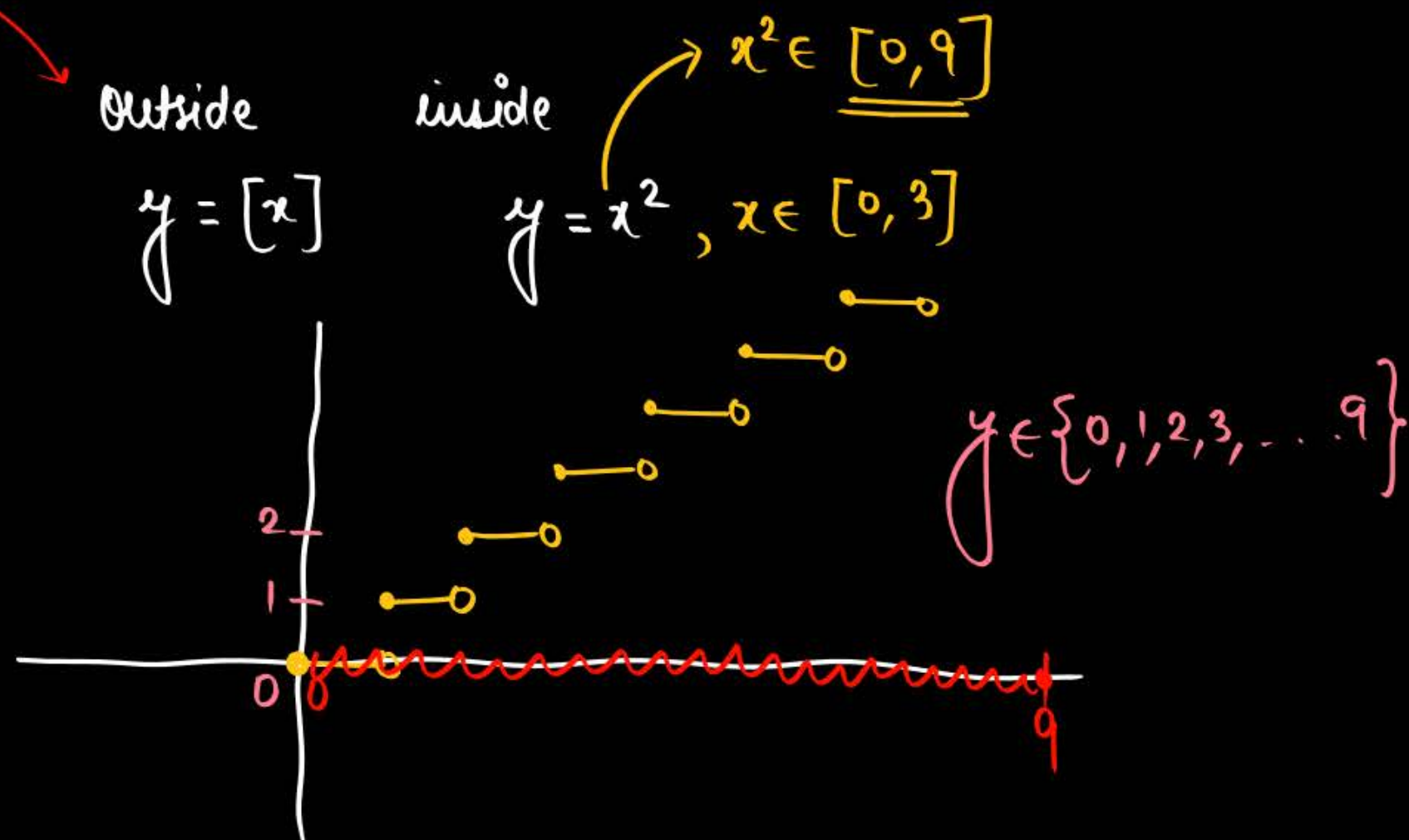
1. $f(x) = [\sin x] \rightarrow y \in \{-1, 0, 1\}$

2. $f(x) = [x^2], x \in [0, 3]$

3. $f(x) = \left[\frac{2}{x}\right], x \in [3, 4)$ $\rightarrow y \in \{0\}$

4. $f(x) = x + \{x\}, x \in [-1, 2)$

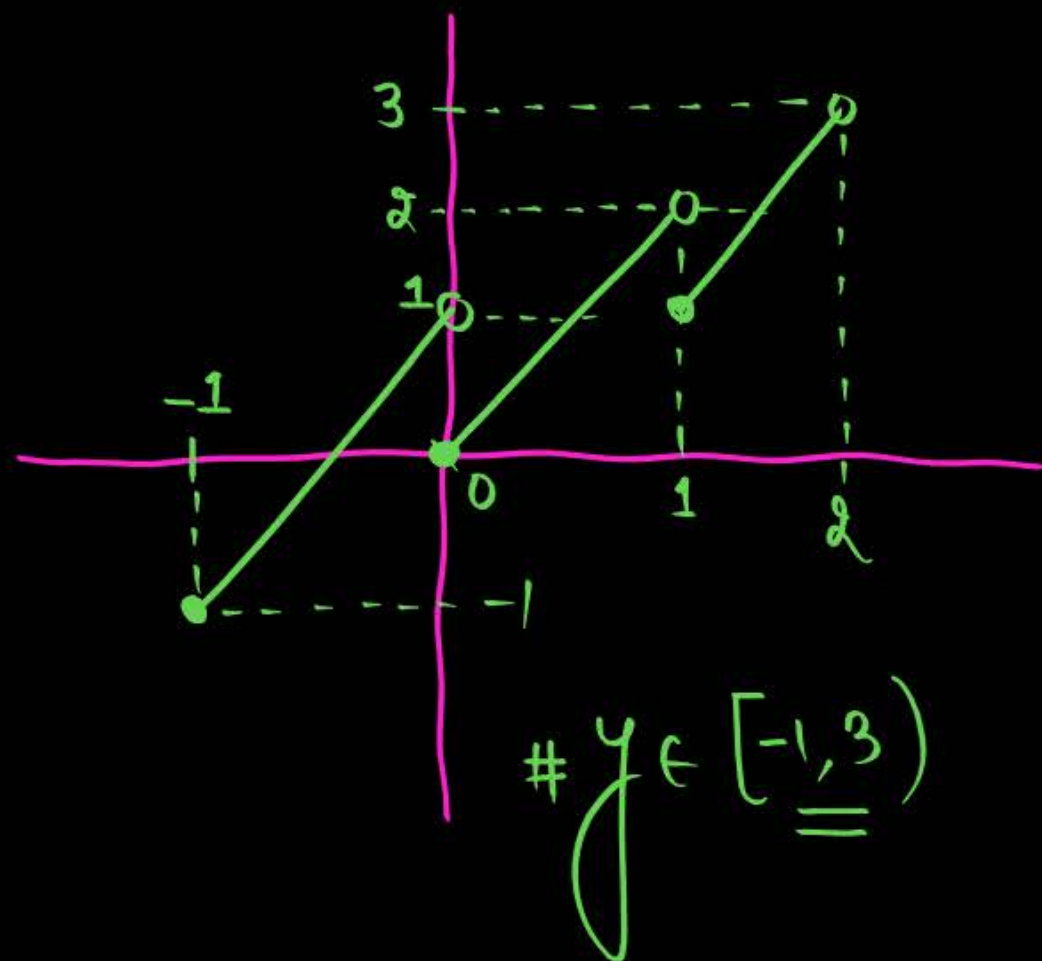
③ $x \in [3, 4)$
 $\frac{1}{x} \in \left(\frac{1}{4}, \frac{1}{3}\right]$
 $\frac{2}{x} \in \left(\frac{1}{2}, \frac{2}{3}\right]$
 $\left[\frac{2}{x}\right] = 0$



④ $y = x + \{x\}, x \in [-1, 2)$

$$y = x + x - [x]$$

$y = 2x - [x], x \in [-1, 2)$



$$\begin{aligned} y &= 2x + 1, & x \in [-1, 0) & \rightarrow 2x + 1 \in [-1, 1) \\ &= 2x, & x \in [0, 1) & \rightarrow 2x \in [0, 2) \\ &= 2x - 1, & x \in [1, 2) & \rightarrow 2x - 1 \in [1, 3) \end{aligned}$$

DIBY-133

$$x \in (1, 3)$$

Let $f: (1, 3) \rightarrow R$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\leq x$. Then the range of f is: [JEE Mains-2020]

(A) $\left(\frac{3}{4}, \frac{4}{5}\right)$

(B) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$

(C) $\left(\frac{2}{5}, \frac{3}{5}\right] \cup \left(\frac{3}{4}, \frac{4}{5}\right)$

(D) $\left(\frac{2}{5}, \frac{4}{5}\right]$

[Ans.: (B)]

$$f(x) = \frac{x[x]}{1+x^2}$$

$$= \frac{x}{1+x^2},$$

$$= \frac{2x}{1+x^2},$$

$x \in (1, 2) \rightarrow \frac{dy}{dx} < 0 \rightarrow f(x) \downarrow$
 $y \in \left(\frac{2}{5}, \frac{1}{2}\right)$

$x \in [2, 3) \rightarrow \frac{dy}{dx} < 0 \rightarrow f(x) \downarrow$
 $y \in \left(\frac{3}{5}, \frac{4}{5}\right]$

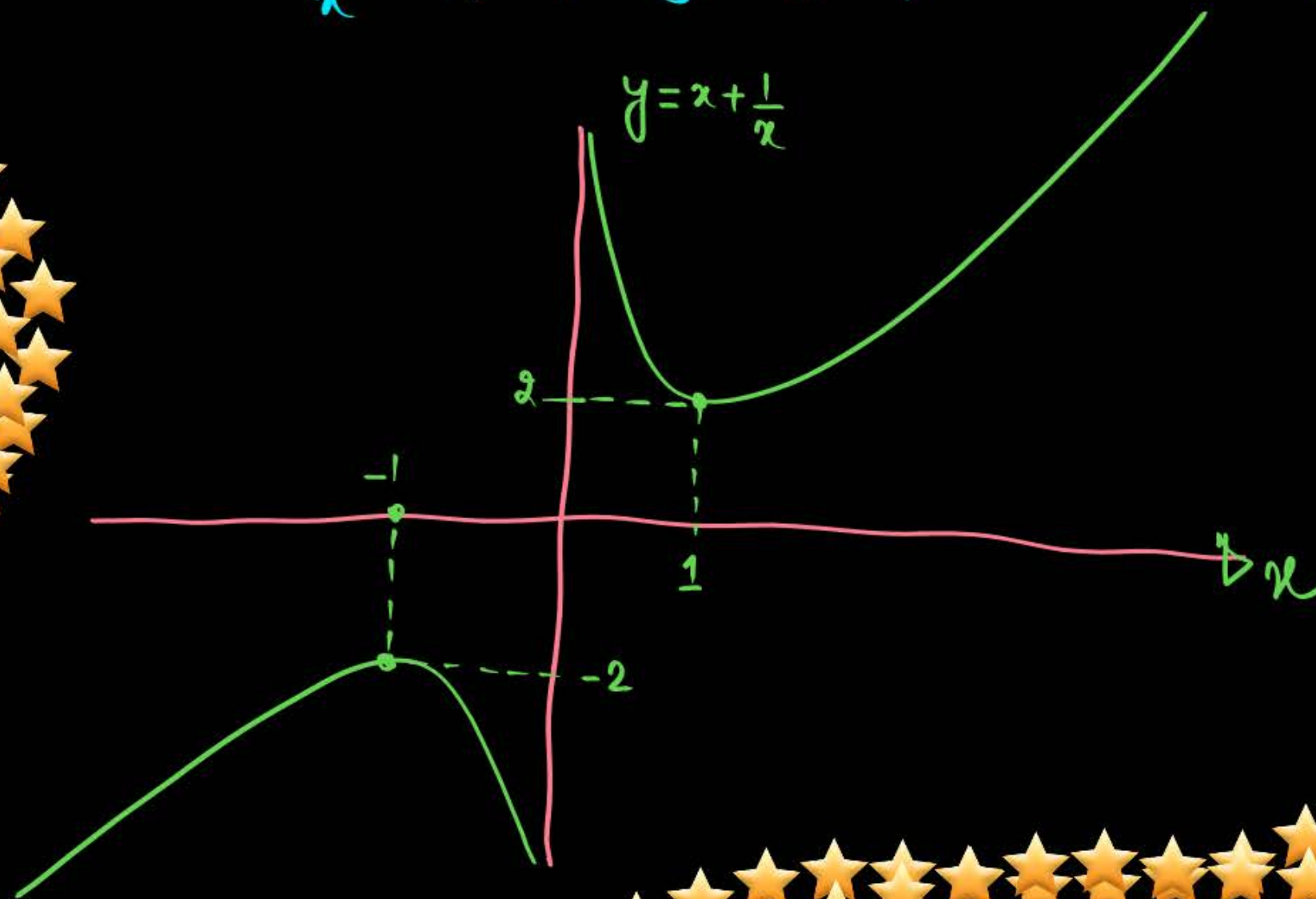
$$y = \frac{2x}{1+x^2}$$

$$\frac{dy}{dx} = \frac{2((1+x^2)1 - (x)(2x))}{(1+x^2)^2} = \frac{2(1-x^2)}{(1+x^2)^2} = 0$$

$x = \pm 1$

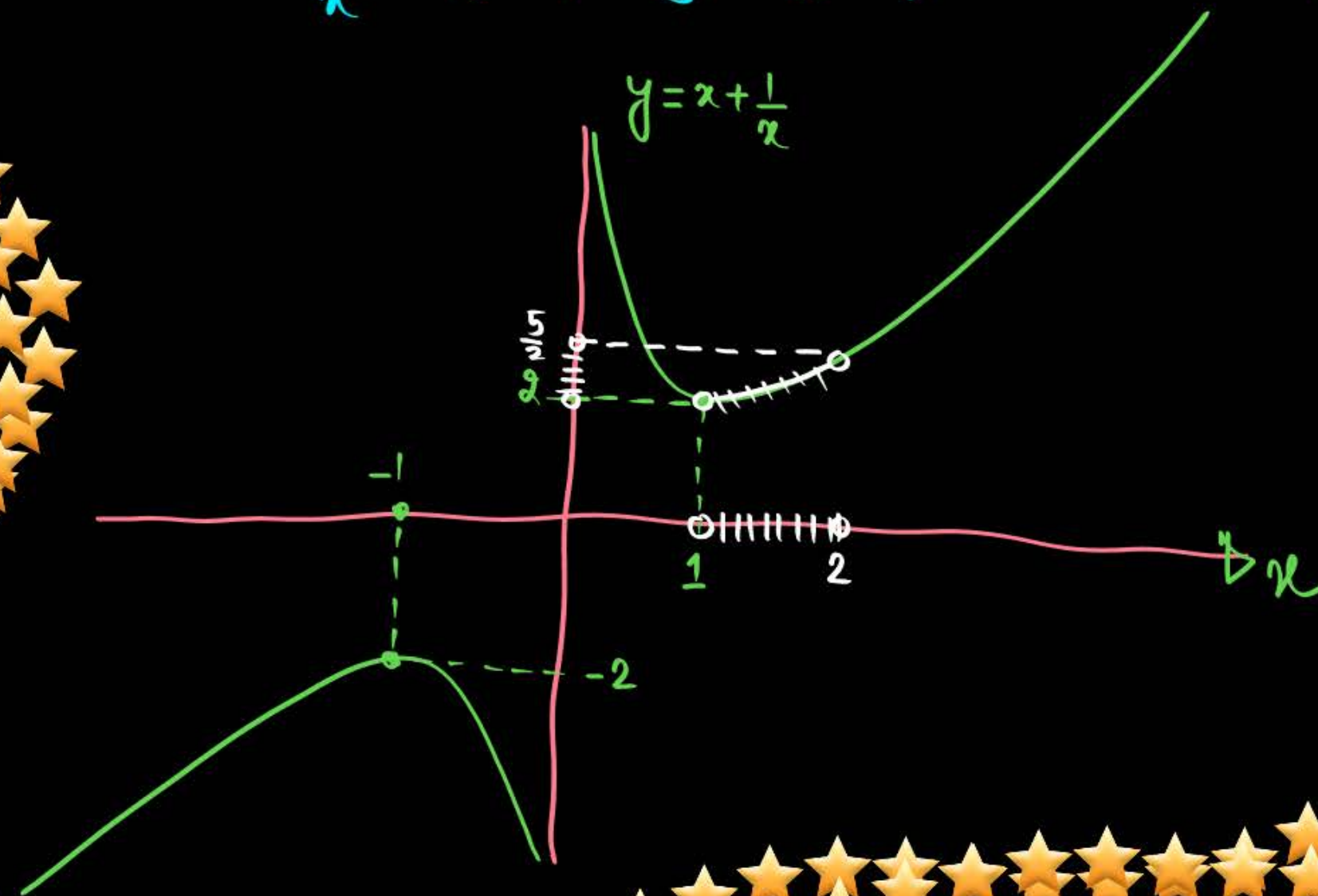
Note:-

$$x + \frac{1}{x} \in (-\infty, -2] \cup [2, \infty)$$



Note:-

$$x + \frac{1}{x} \in (-\infty, -2] \cup [2, \infty)$$



$$\# x \in (1, 2)$$

$$\frac{x}{1+x^2} = \frac{1}{x + \frac{1}{x}}$$

$$x + \frac{1}{x} \in \left(2, \frac{5}{2}\right)$$

$$\frac{1}{x + \frac{1}{x}} \in \left(\frac{2}{5}, \frac{1}{2}\right)$$

Question [JEE Mains-2023]



If the domain of function $f(x) = \frac{[x]}{1+x^2}$, where $[x]$ is greatest integer $\leq x$, is $[2, 6)$, then its range is

- ☒ A $\left(\frac{5}{26}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$
- ☒ B $\left(\frac{5}{26}, \frac{2}{5}\right]$
- ☒ C $\left(\frac{5}{37}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$
- ☒ D $\left(\frac{5}{37}, \frac{2}{5}\right]$

Handwritten solution:

$$f(x) = \frac{[x]}{1+x^2}$$

For $x \in [2, 3)$, $f(x) = \frac{2}{1+x^2} \in \left(\frac{2}{25}, \frac{2}{5}\right]$

For $x \in [3, 4)$, $f(x) = \frac{3}{1+x^2} \in \left(\frac{3}{17}, \frac{3}{10}\right]$

For $x \in [4, 5)$, $f(x) = \frac{4}{1+x^2} \in \left(\frac{4}{26}, \frac{4}{17}\right]$

For $x \in [5, 6)$, $f(x) = \frac{5}{1+x^2} \in \left(\frac{5}{37}, \frac{5}{26}\right]$

Combining all intervals, the range is $\left(\frac{5}{37}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$.

$$y = \frac{k}{1+x^2}, \quad x \in [k, k+1)$$

$$\quad \quad \quad \left(x^2 \in [k^2, (k+1)^2) \right)$$

$$x^2 + 1 \in [k^2 + 1, (k+1)^2 + 1)$$

$$\frac{k}{x^2 + 1} \in \left(\frac{k}{(k+1)^2 + 1}, \frac{k}{k^2 + 1} \right]$$

Challenger Question



Find the number of elements contained in the range of the function

$$f(x) = \left[\frac{x}{6} \right] \left[\frac{-6}{x} \right], \forall x \in (0, 30] \text{ (where } [.] \text{ denotes greatest integer function)}$$

Ans.



EQUAL & IDENTICAL FUNCTION



Two functions f and g are said to be equal if:

- (i) Domain of f = Domain of g**
- (ii) Range of f = Range of g**
- (iii) $f(x) = g(x) \forall x$ in their domain.**

or

Dono ka graph same hona chahiye

Question

$x \in \mathbb{R} - \{n\pi\}$

Find which of following pair is of identical function :

(i) $f(x) = \operatorname{cosec} x, g(x) = \frac{1}{\sin x} \rightarrow \text{identical}$

(ii) $f(x) = \sin x, g(x) = \frac{1}{\operatorname{cosec} x} \rightarrow \text{not identical}$

(iii) $f(x) = \sqrt{\frac{1 - \cos 2x}{2}}, g(x) = \sin x \rightarrow \text{not id.}$

(iv) $f(x) = \frac{1}{|x|}, g(x) = \sqrt{x^{-2}} \rightarrow \text{identical}$

(v) $f(x) = \ln(x+2) + \ln(x-3), g(x) = \ln(x^2 - x - 6) \rightarrow \text{non-id.}$

(vi) $f(x) = \frac{1}{1 + \frac{1}{x}}, g(x) = \frac{x}{1+x} \rightarrow \text{non-id.}$

(vii) $f(x) = x|x|, g(x) = x^2 \operatorname{sgn}(x) \rightarrow \text{identical}$

$$= x^2, x > 0$$

$$= 0, x = 0$$

$$= -x^2, x < 0$$

$$= x^2, x > 0$$

$$= 0, x = 0$$

$$= -x^2, x < 0$$

$$f(x) = \frac{1}{|x|} \rightarrow \mathbb{R} - \{0\}$$

$$\# g(x) = \sqrt{\frac{1}{x^2}}$$

FUNCTIONS

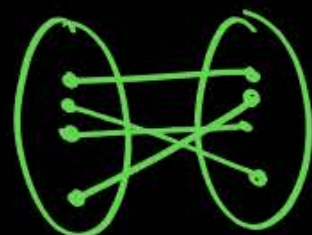
CLASS-XII PORTION



CLASSIFICATION OF FUNCTIONS

INJECTIVE

One-One

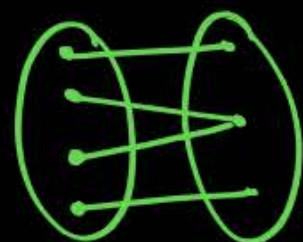


for different x ,
we get diff. values
of y

ex: $y = \ln x, e^x$

NON-INJECTIVE

Many-one



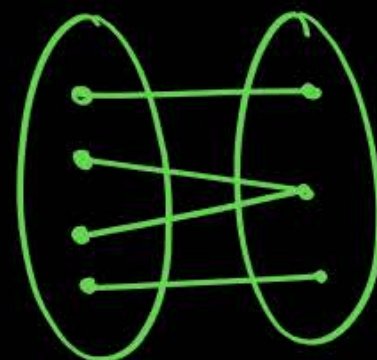
for two or
more values of
 x we get same
 y

ex: $y = \text{all T.F's,}$
 $\{x\}, [x] \dots$
 $|x|, \text{sgn}(x) \dots$

SURJECTIVE

Onto

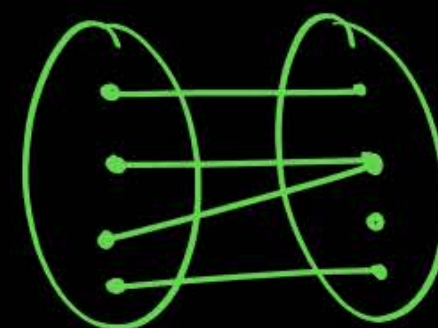
Range = codomain



NON-SURJ.

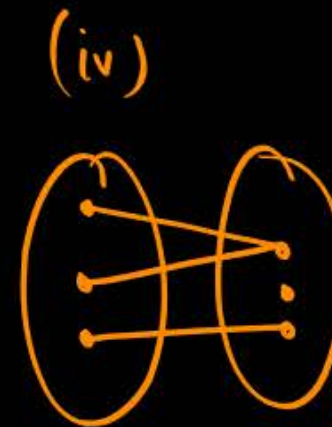
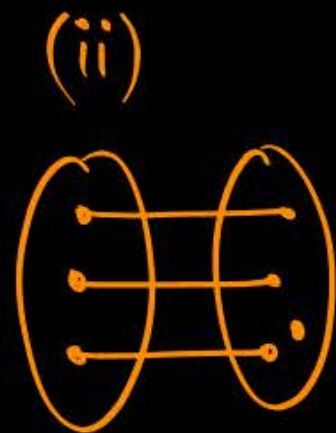
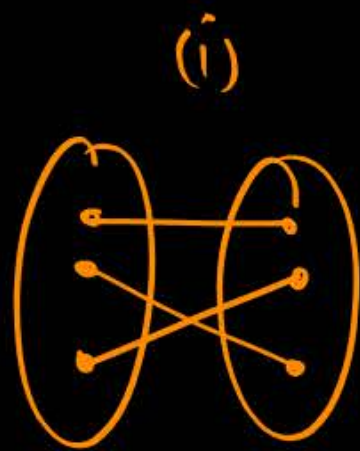
Into

Range $\neq$ Codomain



IMPORTANT NOTE

- (i) **One-One Onto** (Injective and Surjective) ($I \cap S$)
- (ii) **One-One Into** (Injective and Non-Surjective) ($I \cap \bar{S}$)
- (iii) **Many-One Onto** (Surjective but Non-Injective) ($\bar{I} \cap S$)
- (iv) **Many-One Into** (Non-Surjective & Non-Injective) ($\bar{I} \cap \bar{S}$)





HOW TO CHECK ONE-ONE OR MANY-ONE??

M-I: Graphical Method: Any line parallel to X-axis cuts the graph at 2 or more than 2 points, then function is many-one.

M-II: Using $\frac{dy}{dx}$:

If sign doesn't change $\rightarrow$ One-one.

+ always
- always
+ / 0
- / 0

+ $\rightarrow$ -
- $\rightarrow$ +

If sign changes $\rightarrow$ Many-One

M-III: By Observation: (Multiple roots) or (for different 'x' getting same y)

M-IV: Theoretical:

Solve $f(x_1) = f(x_2)$ & analyse. (Boards!)

Bumper Question



Check the following functions as one-one or many-one :

- (i) $f(x) = e^x \rightarrow (1-1)$
- (ii) $f(x) = \{x\} \rightarrow (\text{Many-one})$
- (iii) **All trigonometric functions** $\rightarrow (\text{Many-one})$
- (iv) $f(x) = \text{sgn}(x) \rightarrow (\text{Many-one})$
- (v) **Constant function** $\rightarrow (1-0)$
- (vi) $f(x) = \ln(x) \rightarrow (1-1)$
- (vii) **Even degree polynomial** $\rightarrow \text{Many-one.}$
- (viii) **Odd degree polynomial** $\rightarrow (\text{can't say until func. is given})$
- (ix) **Identity function** $\rightarrow (1-1)$
- (x) $f(x) = \frac{x^2+1}{x} = x + \frac{1}{x} \rightarrow (\text{Many-one})$



Homework



Re-attempt all good Questions of Class

DIBY

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians